

Supplementary Information for “Trapping alkaline earth Rydberg atoms in optical tweezer arrays”

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I. CALCULATION OF RYDBERG TRAPPING POTENTIALS

To calculate the potential seen by a Rydberg atom in an optical tweezer, we compute the energy as a function of the position of the atomic nucleus $\vec{R}$. To make the calculation tractable, we restrict our calculation to states with electron configuration $6snl$, and make the approximation that the inner $6s$ and outer nl electrons can be treated separately [1].

The inner electron polarizability is dominated by the Yb^+ ion $6s - 6p$ transitions. For linearly polarized trapping light, there is only a scalar polarizability, which has been computed at 532 nm to be $\alpha_c = 96$ atomic units (a.u.) for the $^2S_{1/2}$ ground state [2], giving rise to a potential $U_c = -\frac{1}{2\epsilon_0 c} \alpha_c I$. α_c is around 35% of the Yb^0 1S_0 polarizability (275 a.u. [3] or 226 a.u. [4]), indicating that suitably deep traps can be reached with the same powers used to trap ground states. Intriguingly, it is also very close to the calculated polarizability of the metastable Yb^0 $6s6p$ 3P_0 level (95 a.u. [3] or 91 a.u. [4]), which may enable magic-wavelength trapping [1] of high- n Rydberg states and the upper clock state.

The outer, Rydberg electron experiences a spatially-dependent ponderomotive potential proportional to the intensity of the trapping light:

$$U_r = -\frac{1}{2\epsilon_0 c} \alpha_p \int d^3\vec{r} |\psi(\vec{r})|^2 I(\vec{r} + \vec{R}). \quad (\text{S1})$$

Here, $\alpha_p = -e^2/m_e\omega^2$ is the ponderomotive polarizability, $\vec{r}$ is the coordinate of the electron with respect to the nucleus at $\vec{R}$, and $\psi(\vec{r})$ is the electronic wavefunction. This integral is the expectation value of the operator I for the electron wavefunction. Its evaluation is simplified if $I(r)$ can be expanded in irreducible tensor operators; then, the Wigner-Eckart theorem allows U_r to be expressed for any state in terms of angular factors and fewer than $L + 1$ radial integrals. For a lattice, this expansion can be done analytically [5, 6]; however, for a tweezer or other, arbitrary potential it must be done numerically. Specifically, we seek an expansion in spherical harmonics centered on the nuclear coordinate $\vec{R}$:

$$I(\vec{r} + \vec{R}) = \sum_{kq} I_q^{(k)} = \sum_{kq} f_{kq}(r; \vec{R}) C_q^{(k)}(\hat{r}). \quad (\text{S2})$$

Here, $C_q^{(k)}(\hat{r}) = \sqrt{\frac{4\pi}{2k+1}} Y_q^{(k)}(\hat{r})$ is the normalized spherical harmonic and $r = |\vec{r}|$ and $\hat{r} = \vec{r}/r$, and $f_{kq}(r)$ are the coefficients to be found representing I . These functions, which have $\vec{R}$ as a parameter, can be computed by exploiting the orthonormality of the spherical harmonics as:

$$f_{kq}(r; \vec{R}) = \sqrt{\frac{2k+1}{4\pi}} \iint d\Omega I(\vec{R} + \vec{r}) Y_q^{(k)}(\theta, \phi). \quad (\text{S3})$$

Here, the angular integration is performed with respect to $\vec{r}$.

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With the potential decomposed in this way, we can evaluate the matrix element between arbitrary states with quantum numbers $n'l'm'$ and nlm using the Wigner-Eckart theorem:

$$\langle n'l'm'|I|nlm\rangle = \sum_{kq} \langle n'l'm'|I_q^{(k)}|nlm\rangle = \sum_{kq} (-1)^{l'-m'} \begin{pmatrix} l' & k & l \\ -m' & q & m \end{pmatrix} \langle n'l' || I_q^{(k)} || nl \rangle. \quad (\text{S4})$$

The reduced matrix element is given by

$$\langle n'l' || I_q^{(k)} || nl \rangle = (-1)^{l'} \sqrt{(2l'+1)(2l+1)} \begin{pmatrix} l' & k & l \\ 0 & 0 & 0 \end{pmatrix} \int dr r^2 R_{n'l'}(r) R_{nl}(r) f_{kq}(r; \vec{R}), \quad (\text{S5})$$

Here, the term in parentheses is a Wigner $3j$ symbol, and R_{nl} is the radial wavefunction of the Rydberg electron. The potential for a state nlm is:

$$U_r = -\frac{1}{2\epsilon_0 c} \alpha_p \langle nlm|I|nlm\rangle. \quad (\text{S6})$$

This calculation describes the action of U_r in the nlm basis, but the Rydberg states of alkali and alkaline atoms have significantly resolved fine structure splittings. Therefore, these expressions must be expanded in the appropriate basis. Before proceeding further, we note several properties of the potential that are already apparent. First, the contribution of odd- k terms vanishes between states of the same l . Second, $f_{kq} = 0$ when $q \neq 0$ for $\vec{R}$ on the z -axis, since the potential is cylindrically symmetric. Lastly, only terms with $k \leq l$ have non-vanishing contributions, in order to satisfy the conservation of angular momentum (here, l is the total angular momentum, but in the fine structure basis, this will be replaced by $j = l + s$).

Taken together, these allow the potential U_r for low- l states at $\vec{R} = 0$ to be evaluated from a small number of f_{kq} , with k even and $q = 0$. At high- l (i.e., circular states) it appears that a large number of f_{kq} contribute to the potential. However, the radial matrix elements and angular coefficients of high- l states decay rapidly with k , fundamentally as a consequence of the fact that a gaussian beam does not have significant angular momentum, and we find that truncating the calculation to the lowest few k is an excellent approximation.

Additionally, we note that some of the off-diagonal matrix elements $\langle n'l'm'|I|nlm\rangle$ are of similar magnitude to the diagonal elements, and in this sense Eq. S6 is only the first order term in a perturbative calculation of the energy shift. For the trap depths considered here, the total ponderomotive potential is smaller than the spacing between all of the unperturbed energy levels (including m levels, as we apply a 4.5 G magnetic field), such that the next order terms do not change the potential significantly. However, in the absence of external fields, the off-diagonal terms in m can be significant away from the z -axis. More generally, the terms between different l, m may be exploited to drive transitions between Rydberg states in an intensity-modulated beam, where the modulation is resonant with the energy difference between the initial and final states. This could be particularly useful in connection with high-order Laguerre-Gauss beams to create large k terms to efficiently excite circular states (following Ref. [7]) or drive transitions between circular states. Unlike two-photon electric dipole transitions, which are limited to $k \leq 2$, the ponderomotive potential can drive many angular momentum quanta in a single step.

Lastly, we note that the $\vec{R}$ dependence of U_r is not the same as $I(\vec{R})$, because of spatial averaging by the extended Rydberg electron wavefunction. In this work, we consider only the trap depth $U_r(\infty) - U_r(0) = -U_r(0)$. From numerical evaluations of $U_r(\vec{R})$, we have observed that the trap frequencies are slightly higher than would be predicted from the depth alone, since the spatial averaging washes out the (repulsive) ponderomotive potential.

We now consider the evaluation of the reduced matrix elements separately for alkali atoms and alkaline earth atoms.

A. Alkali atoms

To compute the potential for alkali atoms, the matrix elements are needed in the spin-orbit (fine structure) basis: $\langle nsljm|I|nsljm\rangle$. As before, we start with the Wigner-Eckart theorem:

$$\langle nsljm|I_q^{(k)}|nsljm\rangle = (-1)^{j-m} \begin{pmatrix} j & k & j \\ -m & q & m \end{pmatrix} \langle nslj || I_q^{(k)} || nslj \rangle. \quad (\text{S7})$$

Then, we reduce the matrix element again to one only acting on l [8]:

$$\langle nslj || I_q^{(k)} || nslj \rangle = (-1)^{s+l+j+k} (2j+1) \left\{ \begin{matrix} l & j & s \\ j & l & k \end{matrix} \right\} \langle nl || I_q^{(k)} || nl \rangle. \quad (\text{S8})$$

The reduced matrix element here is the same as Eq. (S5), and the coefficients are tabulated in Table S1.

B. Alkaline earth atoms

In the case of divalent Yb, we are interested in calculating the trapping potential for Rydberg states with term symbols $^{2S+1}L_J$ and $6snl$ electronic configurations. The LS-coupled basis is a close approximation to the true eigenbasis: in Yb the measured single-triplet mixing arising from spin-orbit coupling in high- n 3P_1 states is approximately 6% [9], and this effect is presumably smaller in lighter alkaline earth atoms such as Sr. The ponderomotive potential only acts on the outer electron, so we reduce the matrix elements to account for this.

Using the Wigner-Eckart theorem, we calculate the diagonal matrix elements between LS-coupled states with S, L, J denoting the total electronic spin, orbital angular momentum and overall angular momentum, and M the z -projection of J :

$$\langle nSLJM || I_q^{(k)} || nSLJM \rangle = (-1)^{J-M} \begin{pmatrix} J & K & J \\ -M & q & M \end{pmatrix} \langle nSLJ || I_q^{(k)} || nSLJ \rangle. \quad (\text{S9})$$

Then we reduce the matrix element further to one acting on total L :

$$\langle nSLJ || I_q^{(k)} || nSLJ \rangle = (-1)^{L+S+J+k} (2J+1) \left\{ \begin{matrix} L & J & S \\ J & L & k \end{matrix} \right\} \langle nL || I_q^{(k)} || nL \rangle. \quad (\text{S10})$$

Now, we would like to reduce $\langle nL || I_q^{(k)} || nL \rangle$ to a matrix element on the outer electron alone. However, since we are only interested in states where the inner electron is in $6s$ ($l_i = 0$), we have the situation that $L = l_i + l_o = l_o$ and we can just replace L with l_o (the outer electron orbital angular momentum) in the reduced matrix element in Eq. (S10). Therefore, the final result for the ponderomotive potential for alkaline Rydberg states of the form $msnl$ is the same as for alkali atoms [Eq. (S8)], evaluated with the *total* angular momentum quantum numbers $SLJM$ instead of those for the Rydberg electron alone, $sljm$.

C. Numerical evaluation of radial integrals

To evaluate the potential numerically, the radial integrals must be performed. $f_{kq}(r)$ is computed using the gaussian solution to the paraxial wave equation [10]. In the literature, several different approximations have been employed to find Rydberg wavefunctions for the computation of matrix elements using experimentally determined quantum defects, including Coulomb functions and numerical integration of the Schrödinger equation. In this work, we observe that the reduced matrix elements $\langle nl || I_q^{(k)} || nl \rangle$ depend only on the square modulus of the wavefunction and vary slowly with n and l . Therefore, we compute the reduced matrix elements for integer n using hydrogen radial wavefunctions R_{nl} , and interpolate between them to compute the effective matrix element for $n^* = n - \delta_{nSLJ}$, where δ_{nSLJ} is the measured quantum defect.

D. Angular dependence

Since the n and l dependence of the radial integrals is small, the variation of the trapping potential between nearby Rydberg states arises primarily from the angular factors in Eq. (S9) - (S10). These are tabulated for alkali and alkaline earth atoms in Table S1. In the alkali case, states of the same j have the same angular factors and therefore approximately the same trapping potential [11]. In the alkaline case this is no longer true; however, the states 1S_0 , 3S_1 and 3P_0 have the same (purely scalar) potential.

Term	$k = 0$	$k = 2$	$k = 4$
alkali $^2S_{1/2}$	1	0	0
$^2P_{1/2}$	1	0	0
$^2P_{3/2}$	1	1/5	0
$^2D_{3/2}$	1	1/5	0
$^2D_{5/2}$	1	8/35	2/21
alkaline 1S_0	1	0	0
3S_1	1	0	0
1P_1	1	2/5	0
3P_0	1	0	0
3P_1	1	-1/5	0
3P_2	1	1/5	0
1D_2	1	2/7	2/7
3D_1	1	1/5	0
3D_2	1	1/7	-4/21
3D_3	1	8/35	2/21

TABLE S1. Angular factors in U_r , expressed as the coefficient of the radial integral involving f_{k0} for the $m = 0$ state.

II. YB 3S_1 SERIES SPECTROSCOPY

We measure the energies of the $^{174}\text{Yb } 6sns \ ^3S_1$ series using MOT depletion spectroscopy [12, 13]. First, we load a MOT on the 1S_0 to 3P_1 transition in Yb. After compressing the MOT to $\sim 200 \mu\text{m}$ by ramping the power and detuning, we image the MOT on a camera while exposing it to a UV laser with a wavelength near 308 nm, with a beam diameter of approximately 1 mm, and a power of approximately 5 mW. Exciting atoms from 3P_1 to a Rydberg state reduces the MOT fluorescence amplitude in a subsequent fluorescence image. As detailed in the main text, we generate UV light at 308 nm by summing the output of a Ti:Sapphire laser at 1016 nm with a 1565 nm fiber laser and doubling the 616 nm output light in a resonant cavity. To perform high resolution spectroscopy, we change the frequency of the 1016 nm light and measure the frequency of the 616 nm output on a wavemeter with 2 MHz (3σ) accuracy (Toptica WS8-2). The wavemeter is calibrated with the $^{174}\text{Yb } ^1S_0$ to 3P_1 transition frequency (539386602.225 MHz). A typical spectrum is shown in Fig. S1a for $n = 49$.

Table S3 presents the measured energies for the 3S_1 series. The high resolution wavemeter gives an accuracy of 4 MHz in UV laser frequency. To determine the absolute energy of the levels, we add the $^{174}\text{Yb } ^1S_0$ to 3P_1 transition frequency, determined from the measured $^{171}\text{Yb } ^1S_0 F = 1/2$ to $^3P_1 F = 3/2$ frequency (539390406.833 MHz) [14], along with the isotope and hyperfine shifts from [15].

To determine the ^{174}Yb ionization energy, we fit the measured energies for $n < 65$. Fitting the energies to the typical function $E_I - \frac{Ry}{(n-\delta)^2}$, however, results in significant deviations at high n . We find that the measured energies are better fit by the model $E_I - \frac{Ry}{(n-\delta)^2} + a(n-\delta)^7$, which accounts for the expected n^{*7} dependence of Stark shifts due to residual electric fields. As noted in the text, we have intra-vacuum electrodes that allow us to zero the electric field in the region of the atoms. However, changing the potential on the electrodes results in an extremely long (10-30 minutes) settling time for the electric field seen by the Rydberg atoms, which we believe arises from significant field penetration into the glass and slow redistribution of charges there. In practice, this makes finding and maintaining a precise null difficult. We note that in more recent experiments (including those in the main text), the field settling time is shortened dramatically by illuminating the glass cell with a 1 W UV LED at 365 nm (Ref. 27 of [16]). We believe the UV light ionizes charge traps in the glass cell, resulting in much faster equilibration of the charge distribution when the electrode potentials are changed. This allows the suppression of stray electric fields at the level of several mV/cm. The resulting offset potentials are stable for weeks, which will allow more precise study of the high- n spectroscopy in the future, although we were not able to repeat the high-precision spectroscopy because the necessary high-resolution wavemeter was no longer available to us.

The sign of the shifts (towards higher energy, $a > 0$) is consistent with the measured sign of the quadratic Stark coefficient for the 3S_1 series. The magnitude of the shifts (22 MHz at $n=100$) is consistent with our estimated precision in setting the field null (which was done at $n = 80$ before beginning data acquisition at $n = 100$ and stepping down in n). The ionization energy extracted from this fit is $E_I = 50443.07047(4) \text{ cm}^{-1}$, within 0.00006 cm^{-1} (2 MHz) of the value reported in Ref. [17] (Ry is the Rydberg constant for Yb, $109736.96959 \text{ cm}^{-1}$).

Using this ionization energy, we determine the measured quantum defects for $n = 28$ to $n = 100$, shown in Fig.

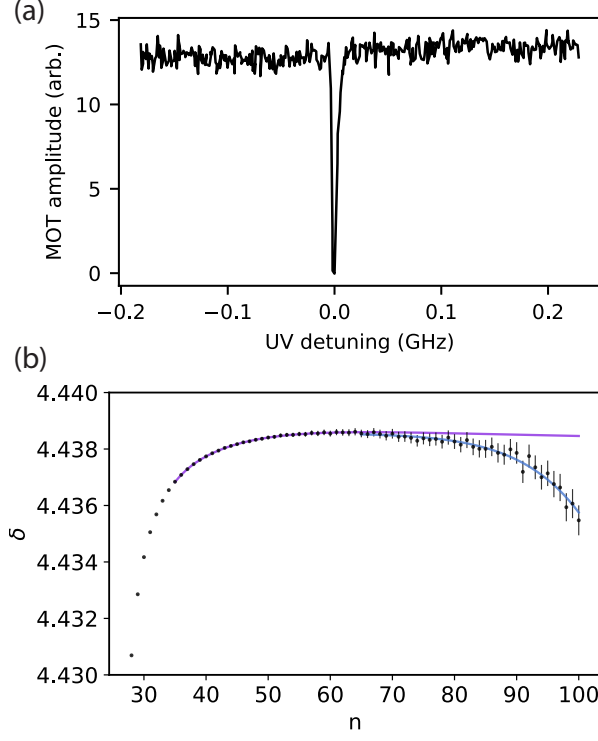


FIG. S1. (a) Example spectrum for the 3S_1 Rydberg line at $n = 49$. (b) Measured quantum defects of the 3S_1 Rydberg series vs. principal quantum number. Error bars show uncertainties in the defects from the 4 MHz uncertainty in the measured UV frequencies. The purple line is a fit to an extended Rydberg-Ritz model for $35 < n < 65$. The blue line results from a fit to $E_I - \frac{R_y}{(n-\delta)^2} + a(n-\delta)^7$. The $(n-\delta)^7$ term accounts for shifts due to residual electric fields in our glass cell.

S1b. To obtain an empirical model for the defects at low n , we fit them to the extended Rydberg Ritz formula:

$$\delta(n) = \delta_0 + \frac{\delta_2}{(n-\delta_0)^2} + \frac{\delta_4}{(n-\delta_0)^4} + \dots \quad (\text{S11})$$

The fit parameters for the region $35 < n < 65$ are summarized in Table. S2, and the fitted energies are within our experimental uncertainty in this range.

Fit Parameter	Value
δ_0	4.4382(2)
δ_2	4(1)
δ_4	$-10(4) \times 10^3$
δ_6	$8(4) \times 10^6$
δ_8	$-3(2) \times 10^9$

TABLE S2. Parameters for the fit (Fig. S1b) to the extended Rydberg-Ritz model in Eq. (S11), for $35 < n < 65$.

This measurement of the 3S_1 quantum defect also allows us to estimate the C_6 coefficient describing the van der Waals interaction for this Rydberg series. We use spectroscopic data of the 3P_j quantum defects from [18] (extrapolating the 3P_0 and 3P_1 values from the measured range), and compute the van der Waals coefficient using exact diagonalization with a small basis [19]. In the range $n = 40$ to 80 , we find that the interaction is attractive, with a value of C_6 that is approximately $-1/2$ times the value for the Rubidium. Importantly, this is in contrast to the Yb 1S_0 , which is predicted to have a small, repulsive interaction [20].

This calculation also predicts that certain states in the range $n = 45 - 60$ have significant deviations arising from Förster resonances with the strongly perturbed 3P_0 and 3P_1 series. The precise value of these perturbed C_6 coefficients, as well as which states experience these resonances, cannot be determined precisely from the existing spectroscopic data.

n	Energy (cm^{-1})	n	Energy (cm^{-1})
28	50245.5285	64	50412.1374
29	50261.2497	65	50413.1505
30	50275.1772	66	50414.1147
31	50287.5688	67	50415.0329
32	50298.6401	68	50415.9083
33	50308.5712	69	50416.7432
34	50317.5130	70	50417.5402
35	50325.5925	71	50418.3016
36	50332.9169	72	50419.0294
37	50339.5773	73	50419.7256
38	50345.6515	74	50420.3921
39	50351.2064	75	50421.0303
40	50356.2995	76	50421.6420
41	50360.9806	77	50422.2285
42	50365.2929	78	50422.7914
43	50369.2741	79	50423.3316
44	50372.9574	80	50423.8507
45	50376.3717	81	50424.3495
46	50379.5425	82	50424.8291
47	50382.4925	83	50425.2906
48	50385.2417	84	50425.7348
49	50387.8079	85	50426.1625
50	50390.2071	86	50426.5745
51	50392.4533	87	50426.9718
52	50394.5593	88	50427.3548
53	50396.5366	89	50427.7242
54	50398.3955	90	50428.0809
55	50400.1451	91	50428.4254
56	50401.7940	92	50428.7578
57	50403.3496	93	50429.0794
58	50404.8190	94	50429.3902
59	50406.2082	95	50429.6906
60	50407.5232	96	50429.9813
61	50408.7690	97	50430.2627
62	50409.9505	98	50430.5352
63	50411.0720	99	50430.7988
64	50412.1374	100	50431.0545

TABLE S3. Measured energies of the $^{174}\text{Yb } 6sns \ ^3S_1$ Rydberg series from $n = 28$ to $n = 100$.

III. TRAP-INDUCED LOSS MECHANISMS

state	Lifetime (μs)
$n = 74 \ ^3P_2$	83(5)
$n = 74 \ ^3P_0$	14(4)*
$n = 73 \ ^1D_2$	75(18)
$n = 83 \ ^1D_2$	59(3)
$n = 90 \ ^1D_2$	60(3)
$n = 70 \ ^3S_1$	85(4)
$n = 75 \ ^3S_1$	105(3)
$n = 92 \ ^3S_1$	42(2)

TABLE S4. Summary table of power dependent lifetime studies for various Rydberg states, showing the extrapolated lifetimes with no trap-induced losses. *We do not have power dependence data to quote the extrapolated lifetime for 3P_0 , but give the value for 9 mW trap power.

A. Auto-ionization

The polarizability of the Yb^+ ion core primarily arises from the $6s$ to $6p_{1/2}$ and $6p_{3/2}$ transitions at 369 and 329 nm, respectively. For an optically trapped Yb^+ ion, the finite lifetime of these states ($1/\Gamma = 8$ ns for $6p_{1/2}$, [21]) would give rise to photon scattering at a rate $\Gamma\Omega^2/\Delta^2 = \Gamma U_0/\Delta$, where Ω and Δ are the Rabi frequency and detuning of the trapping laser, and U_0 is the trap depth. In the case of a trapped Yb^0 Rydberg state, the core transitions are between $6snl$ and $6p_jnl$ states (where n, l are the quantum numbers of the outer Rydberg electron). Since the $6p_jnl$ states are above the ionization limit of the $6snl$ series, they can rapidly auto-ionize. In an auto-ionization event, the core electron is de-excited by ejecting the Rydberg electron. The rate of this process decreases as $1/n^{*3}$, since the interaction between the Rydberg and core electron decreases as the former moves farther out. The rate of these events is the same as photon scattering for a Yb^+ ion, with the linewidth Γ replaced by the auto-ionizing linewidth $\gamma'n^{*-3}$, according to the isolated core electron approximation [22].

The auto-ionization rates have been measured for certain $5p_jns$ states in Sr [23] and $6p_jns$ states in Ba (cited in Ref. 21 of [23]) and Yb [24]. All are within the range of $\gamma' = 2\pi \times 10^{14} - 2\pi \times 10^{15} \text{ s}^{-1}$. The rate for the Yb $6p_{3/2}ns$ series has not been measured to the best of our knowledge, but for concreteness we take the measured value for the $6p_{1/2}ns$ series of $\gamma' = 1.2 \times 10^{15} \text{ s}^{-1}$ [24], and assume the $6p_{3/2}ns$ rate is 2 times higher, which is approximately the case for Sr and Ba. From this, we can estimate an auto-ionization rate of about $n^{*-3} 58 \times 10^6 \text{ s}^{-1}$ at typical trap powers (9 mW). This corresponds to a 7 ms lifetime at $n = 75$, which is several orders of magnitude slower than the 3S_1 lifetimes we observe. The rate should be even smaller for higher l states, since the overlap with the core decreases for high l . This is supported by measurements [22, 25] for $l < 7$ and theory predicting an l^{-5} scaling of the autoionization rate for high l [26].

B. Photo-ionization

Another loss mechanism is direct excitation of the Rydberg electron into continuum states, termed photoionization [27, 28]. Calculating this process requires computing matrix elements between bound and free Rydberg states, which requires either model potentials or extrapolation of measured bound state quantum defects into the continuum [29]. The extrapolation quantum defects based on the Rydberg-Ritz model has been studied for alkali atoms [29]; however, given the very strong n -dependence near the ionization threshold observed for the Yb 3P_0 and 3P_1 states [18], a multi-channel quantum defect model would be necessary for a precise calculation of photo-ionization rates for Yb. Furthermore, it is possible to have perturbers above the ionization threshold that would not be evident from the bound state quantum defects.

We instead determine the photoionization rates for a few S , P , and D states by measuring the Rydberg state lifetime as a function of trap power. Here, we assume that all power-dependent loss results from photoionization, since the predicted autoionization rates are very small. Fig. S2a shows an example measurement for the $n = 74$ 3P_2 state with the corresponding decay rates and a fit to $\Gamma = \Gamma_0 + \gamma_{PI}P$, where P is the trap power. Extrapolating the decay rates to zero trap power gives an estimate for the natural Rydberg state lifetimes $1/\Gamma_0$, which are summarized in Table S4. Fig. S2b shows the photoionization cross sections for the measured states, related to the decay rates by $\gamma_{PI}P = \sigma_{PI}I/\hbar\omega$. The measured photoionization cross sections are consistent with zero for the 3S_1 series, but the rates are non-negligible for the 3P_2 and 1D_2 states, resulting in a 15-30% reduction of the trapped lifetimes at typical trap powers. The observed magnitude and low- L dependence of the cross section is similar to previous calculations for Rb [28]. As with auto-ionization, the photoionization rate should decrease for high- L Rydberg states, in this case exponentially with L [30].

IV. 3S_1 LIFETIME AT HIGH n

Given that we observe no trap-induced losses for the 3S_1 series, the measured decrease in the 3S_1 lifetimes at high n is surprising. For alkali atoms, the finite Rydberg state lifetime is a combination of spontaneous emission to much lower- n Rydberg states, and blackbody-induced transitions between nearby Rydberg levels [31]. These rates scale as n^{*-3} and n^{*-2} respectively. Lifetimes of greater than 250 μs have been measured for Rb Rydberg states at $n = 85$ at room temperature [32]. In alkaline earth atoms, an additional loss mechanism is present: configuration interactions can mix Rydberg series attached to different ionization thresholds associated with different core ion states, which results in admixtures of fast-decaying low- n states into the high- n states. Rydberg state energies and lifetimes can be highly irregular near a perturbing resonance. Far from a resonance, however (e.g., at high n), the impact of the perturber is a constant factor reduction of the Rydberg series radiative lifetime [33]. Perturbers are almost certainly

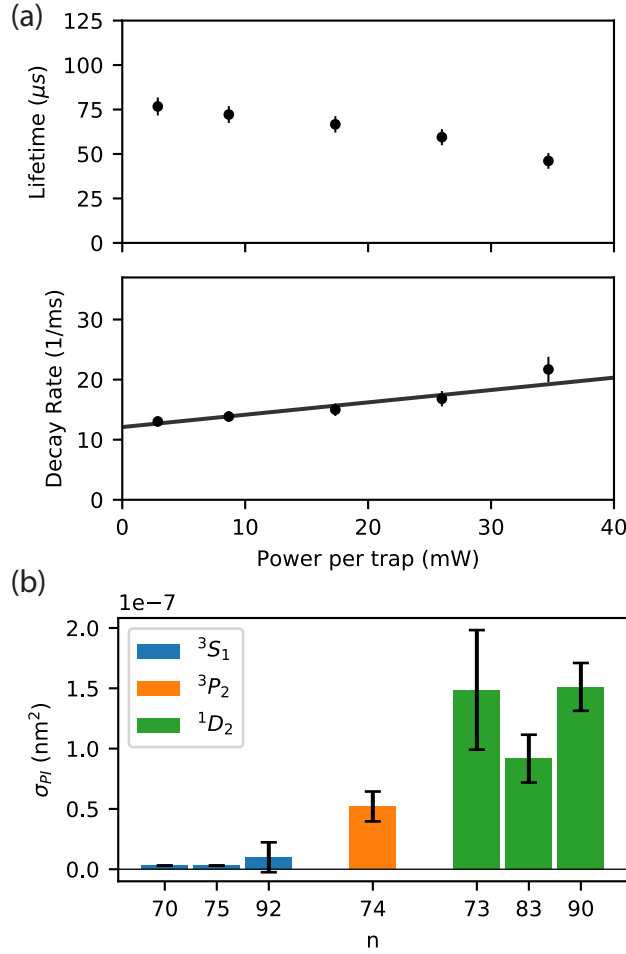


FIG. S2. (a) Rydberg lifetimes vs. trap power for the $n = 74^3P_2$ state. The lower plot shows the corresponding decay rates and a linear fit, the slope of which determines the photoionization cross section σ_{PI} . (b) Measured photoionization cross sections for 3S_1 , 3P_2 , and 1D_2 states at different n . We see no evidence of photoionization for S states.

responsible for the measured, short lifetime of the 3P_0 series (Table S4), as this series is known to be extremely strongly perturbed [18].

Whether series perturbations are responsible for the drop in 3S_1 lifetime at high- n is unclear. While the low- n 3S_1 spectrum (Fig. S1) appears relatively unperturbed (compared to 3P_0 , for example, where the quantum defect varies by nearly 0.5 between $n = 30$ and $n = 80$ [18]), the deviation from constant δ at high- n could reflect an above-threshold perturber, or a technical artifact in the measurement, as discussed in section II. Another possibility is high-frequency electric field noise conducted to the atoms by a set of intra-vacuum electrodes around the atoms. We have simulated the local density of states in the center of the electrodes at microwave frequencies and find low- Q resonances giving rise to Purcell enhancement factors of 2-3 at 7 and 15 GHz, where the strongest transitions lie for $n = 65 - 100$, which could enhance the blackbody transition rate by a similar factor or enhance the coupling of noise at these frequencies. This would also reduce certain high- n lifetimes for other series as well, but we do not have sufficient data on the n -dependence of the P and D series to confirm this. At present, we are unable to experimentally distinguish whether the high- n lifetimes are intrinsic or limited by technical effects.

Lastly, we note that the theoretical lifetime can in principle be extracted from MQDT analysis of the Rydberg spectrum. This has been attempted in some detail for Strontium [33]. In the case of Yb, this is hampered by the absence of complete spectroscopic data, the larger number of perturbers that result from low-lying f -shell excitations in the core (e.g., states of the form $4f^{13}5d6s$), and the fact that the lifetimes of the open f -shell states are not known in many cases. The lifetimes presented in this paper are, to the best of our knowledge, the first reported (or predicted) lifetimes for high- n Yb Rydberg states. Together with additional spectroscopic data, this will aid future analyses of

the complete Yb Rydberg spectrum.

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